

## SEMESTER END EXAMINATIONS – APRIL / MAY 2023

|             |                                                                                        |            |                |
|-------------|----------------------------------------------------------------------------------------|------------|----------------|
| Program     | : <b>B.E :- Common to CSE (CY) / CSE (AI &amp; ML) / (AI &amp; ML) / (AI &amp; DS)</b> | Semester   | : <b>III</b>   |
| Course Name | : <b>Discrete Mathematical Structures</b>                                              | Max. Marks | : <b>100</b>   |
| Course Code | : <b>CI35 / CY35 / AI34 / AD34</b>                                                     | Duration   | : <b>3 Hrs</b> |

### Instructions to the Candidates:

- Answer one full question from each unit.

### UNIT - I

- Define: CO1 (06)
    - Proposition
    - Universal quantifier
    - Tautology and Contradiction.
  - Suppose that the integers 1,2,3,.....,10 are randomly positioned around a circular wheel. Show that by contradiction the sum of same set of consecutive positioned numbers is atleast 15. CO1 (07)
  - Prove by mathematical induction: CO1 (07)
 
$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + 3 \cdot 4 \cdot 5 + \dots + n(n+1)(n+3) = \frac{1}{4}(n)(n+1)(n+2)(n+3) \quad n \geq 1$$
- For the following statements, write converse, inverse and contrapositive CO1 (06)
    - "The universe consists of all integers".
    - "If m divides n and n divides p then m divides p".
  - By constructing truth table show that: CO1 (07)
 
$$[(p \wedge q) \rightarrow r] \leftrightarrow [(p \rightarrow r) \wedge (q \rightarrow r)].$$
  - Write down the following quantified statements in symbolic form. Also negate the statement (v). (i) At least one integer is even. (ii) Some integers are divisible by 3. (iii) Every integer is either even or odd. (iv) If x is even and a perfect square then x is not divisible by 3. (v) If x is odd or is not divisible by 7 then x is divisible by 3. CO1 (07)

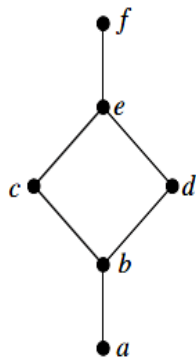
### UNIT - II

- Determine whether the relation R on the set of all people is reflexive, symmetric, antisymmetric, and/or transitive, where  $(a, b) \in R$  if and only if CO2 (06)
    - a is taller than b.
    - a and b were born on the same day.
    - a has the same first name as b.
    - a and b have a common grandparent.
  - Determine whether the following Posets are lattices: CO2 (07)
    - $(\{1, 3, 6, 9, 12\}, |)$
    - $(\{1, 5, 25, 125\}, |)$ .
 Draw Hasse diagram in each case and also verify whether ii) is Boolean algebra.
  - Let m be an integer with  $m > 1$ . Show that the relation CO2 (07)
 
$$R = \{(a, b) / a \equiv b \pmod{m}\}$$
 is an equivalence relation on the set of integers.

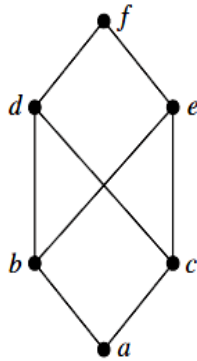
4. a) Determine whether the relation represented by the following matrix is reflexive, irreflexive, symmetric, antisymmetric, and/or transitive. CO2 (06)

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

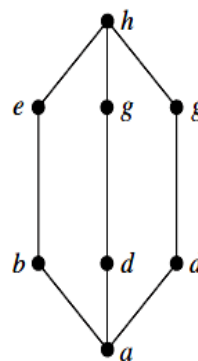
- b) Determine whether the Posets represented by each of the following Hasse diagrams are lattices. CO2 (07)



(a)



(b)



(c)

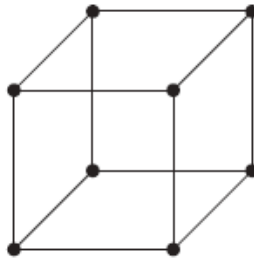
- c) Answer the following questions for the Poset  $(\{2, 4, 6, 9, 12, 18, 27, 36, 48, 60, 72\}, |)$ . CO2 (07)
- Find the maximal elements and minimal elements.
  - Is there a greatest element?
  - Is there a least element?
  - Find all upper bounds of  $\{2, 9\}$ .
  - Find the least upper bound of  $\{2, 9\}$ , if it exists.
  - Find all lower bounds of  $\{60, 72\}$ .
  - Find the greatest lower bound of  $\{60, 72\}$ , if it exists.

### UNIT - III

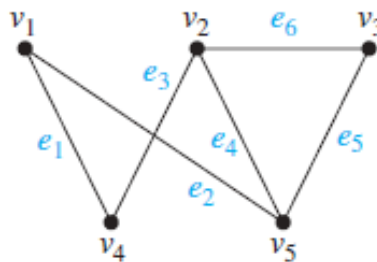
5. a) Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there? CO3 (06)
- b) How many cards must be selected from a standard deck of 52 cards to guarantee that i) at least three cards of the same suit are selected? CO3 (07)
- ii) at least three hearts are selected?
- c) Find an explicit formula for the Fibonacci numbers. CO3 (07)
6. a) How many bit strings of length eight either start with a 1 bit or end with the two bits 00? CO3 (06)
- b) During a month with 30 days, a baseball team plays at least one game a day, but no more than 45 games. Show that there must be a period of some number of consecutive days during which the team must play exactly 14 games. CO3 (07)
- c) Find the solution of the recurrence relation  $a_n = 6a_{n-1} - 9a_{n-2}$  with initial conditions  $a_0 = 1$  and  $a_1 = 6$ . CO3 (07)

UNIT- IV

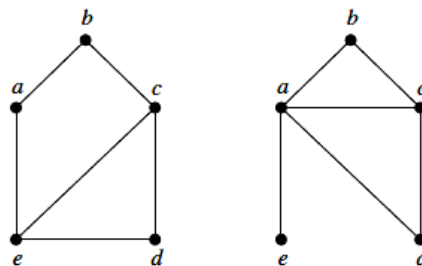
7. a) Define Planar graph and chromatic number. Determine whether the following graph is planar. If so, represent the graph.



- b) Represent the graph shown below with an incidence matrix and adjacency matrix.



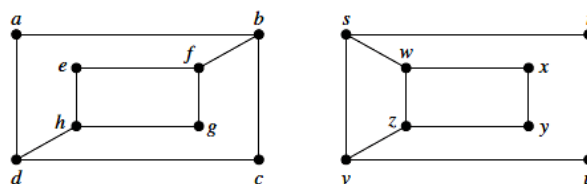
- c) Determine whether the graphs shown below are isomorphic: CO4 (07)



i)

G

H

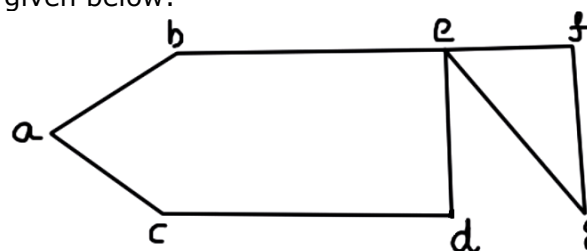


ii)

G

H

8. a) In the graph given below: CO4 (06)



Determine (i) Walk from b to b that is not a trial. (ii) b-d trial that is not a path. (iii) a path from b to d. (iv) a closed walk from b to b that is not a circuit. (v) a circuit from b to b that is not a cycle (vi) a cycle from b to b.

- b) Define: i) Euler path and circuit , and ii) Hamiltonian path and circuit CO4 (07)  
with an example for each.
- c) Determine whether the graphs without loops with the following incidence CO4 (07)  
matrices are isomorphic:

$$d \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

**UNIT - V**

9. a) Define semi group, monoid and group with an example for each. CO5 (06)  
b) Let  $f: \mathbb{R}^+ \rightarrow \mathbb{R}$  defined by  $f(x) = \log(x)$  for any  $x$  belongs to  $\mathbb{R}^+$  with respect CO5 (07)  
to usual addition  $+$  and usual multiplication  $\times$ . Is  $f$  homomorphic ,  
isomorphic or both? Explain.
- c) (i) Define Euler' phi function and give an example. CO5 (07)  
(ii) State Fermat's little theorem. Using this compute  $(128)^{129} \bmod 17$ .
10. a) Let  $G$  be set of all non zero real numbers and let  $a * b = (ab/2)$ . Show that CO5 (06)  
 $(G, *)$  is an abelian group.
- b) If  $G$  is a group. Let  $H = \{a \in G / ag = ga, \forall g \in G\}$ . Prove that  $H$  is a sub CO5 (07)  
group of  $G$ .
- c) Let  $(\mathbb{Z}, +)$  and  $(E, +)$  be two semi groups where  $\mathbb{Z}$  is the set of all integers, CO5 (07)  
 $E$  is the set of all even integers. Define  $f: \mathbb{Z} \rightarrow E$  by  $y = 2x, \forall x \in \mathbb{Z}, y \in E$ .  
Prove that  $f$  is isomorphic.

\*\*\*\*\*